The Radon–Nikodym Theorem

The derivative hiding underneath next-token prediction

One could sum up this whole document in a sentence:

> Whenever one measure respects the null sets of another, it can be written as a weighted integral of it — and that weight, the Radon–Nikodym derivative, is the object a language model's loss is secretly integrating.

The companion note *Next-Token Prediction is Geometry* never names this theorem. It did not need to: in a finite vocabulary the theorem is a triviality. But the moment one asks *what the loss measures*, the derivative $d\nu/d\mu$ is the thing under the integral, and absolute continuity is the constraint that keeps a trained model honest.

Absolute continuity

Fix a measurable space $(X, \mathcal{F})$ — a set X together with a collection $\mathcal{F}$ of subsets we are allowed to measure. A *measure* μ assigns to each $A \in \mathcal{F}$ a size $\mu(A) \in [0, \infty]$, additively over disjoint pieces. It is *σ-finite* if X is a countable union of sets of finite measure — enough room to work piece by piece. Counting measure on a finite set, length on $\mathbb{R}$, and any probability measure are all σ-finite.

Given two measures μ and ν on the same space, say ν is *absolutely continuous* with respect to μ, written $\nu \ll \mu$, when

$$\mu(A) = 0 \implies \nu(A) = 0.$$

In words: wherever μ sees nothing, ν must also see nothing. μ's null sets are a floor that ν is not allowed to build on.

> **Insight** Absolute continuity is a one-directional agreement about *nothing*, not about *something*. It says the two measures agree on which sets are negligible; it says nothing about how they weigh the sets that are not. That asymmetry is exactly why the derivative below runs one way, $d\nu/d\mu$, and need not exist the other way.

The theorem

Theorem (Radon–Nikodym). *Let μ and ν be σ-finite measures on $(X, \mathcal{F})$ with $\nu \ll \mu$. Then there exists a measurable function $f : X \to [0, \infty)$, unique up to equality μ-almost everywhere, such that*

$$\nu(A) \;=\; \int_A f \, d\mu \qquad \text{for every } A \in \mathcal{F}.$$

The function f is the *Radon–Nikodym derivative*, written

$$f = \frac{d\nu}{d\mu}.$$

The content of the theorem is that absolute continuity — a purely set-theoretic condition about null sets — is *exactly* the condition under which ν can be reconstructed as a weighted integral of μ. The weight is the derivative. Nothing weaker than $\nu \ll \mu$ will do, and nothing stronger is needed.

The name is earned, not borrowed. The derivative obeys the laws one expects of a derivative, all holding μ-almost everywhere:

- **Linearity:** $\dfrac{d(\nu_1 + \nu_2)}{d\mu} = \dfrac{d\nu_1}{d\mu} + \dfrac{d\nu_2}{d\mu}$.
- **Chain rule:** if $\nu \ll \mu \ll \lambda$ then $\dfrac{d\nu}{d\lambda} = \dfrac{d\nu}{d\mu}\,\dfrac{d\mu}{d\lambda}$.
- **Reciprocal:** if $\nu \ll \mu$ and $\mu \ll \nu$ (so each respects the other's null sets, written $\mu \equiv \nu$) then $\dfrac{d\nu}{d\mu} = \left(\dfrac{d\mu}{d\nu}\right)^{-1}$.

> **Insight** "Unique up to μ-almost everywhere" is the honest form of uniqueness. The value of f on any μ-null set is invisible to every integral $\int_A f\,d\mu$, so it can be changed freely. The derivative is therefore not a single function but a point in $L^1(\mu)$ — an equivalence class of functions agreeing off the null sets. This is the same shrug that the companion note makes about the manifold being "loose": the object is pinned down exactly as much as it is observable, and no more.

Why the hypothesis is the whole theorem

It is tempting to read $\nu \ll \mu$ as a technicality. It is not; it is the content. Suppose it failed: some set A has $\mu(A) = 0$ but $\nu(A) > 0$. Then no density f could ever reproduce ν, because

$$\int_A f\,d\mu = 0 \quad \text{for } \textit{any } f,$$

since integrating over a μ-null set returns zero no matter how large f is made there. The mass ν places on A is mass that integration against μ is structurally unable to account for. Absolute continuity is precisely the promise that no such orphaned mass exists.

The canonical failure is a point mass. Let μ be Lebesgue measure (length) on $\mathbb{R}$ and $\nu = \delta_0$ the Dirac measure at the origin, $\nu(A) = 1$ if $0 \in A$ and 0 otherwise. The single point $\{0\}$ has length $\mu(\{0\}) = 0$ but $\nu(\{0\}) = 1$. There is no function f with $\int_A f\,d\mu = \delta_0(A)$: one would need f to be zero everywhere except at a single point yet integrate to one, which Lebesgue integration forbids. The Dirac "delta function" is exactly the derivative that the theorem refuses to grant, because its measure is not absolutely continuous.

Three worlds: where the density lives and where it dies

The hypothesis $\nu \ll \mu$ sorts every distribution into one of three regimes, and walking them in order is the quickest way to see what Radon–Nikodym is *for* — and why a discrete model never meets it while a continuous one cannot avoid it.

Discrete: points carry mass

On a finite or countable set, each outcome is an island holding its own probability, $p(x) = P(\{x\})$. The mass function is a self-sufficient object — a real number attached to each point — and every quantity is a sum, $P(A) = \sum_{x \in A} p(x)$. Here the derivative dP/dQ is the elementary ratio $p(x)/q(x)$, and the theorem asserts nothing a calculator cannot: against counting measure every function is a legal density, and absolute continuity $P \ll Q$ can fail only where $q(x) = 0 < p(x)$ — a division by zero visible to the naked eye. This is the codeword world, and Radon–Nikodym sleeps through it.

Absolutely continuous: mass on sets, and a density reborn

On a continuum — $\mathbb{R}$, say — the island picture collapses. Every single point has probability zero, $P(\{x\}) = 0$, so the naive mass function is identically zero and tells you nothing, even as the distribution spreads real probability across intervals, $P([a, b]) > 0$. Probability is no longer a function of *outcomes*; it is a function of *sets*. That is the whole reason measure theory is unavoidable here and absent in the discrete world: the point-function has died, and only the set-function survives.

To recover something point-like, introduce a yardstick — Lebesgue measure λ, "length" — and ask not for the probability *at* x (which is zero) but for the probability *per unit length* near x. That rate is the density,

$$P([a, b]) = \int_a^b f \, d\lambda, \qquad f = \frac{dP}{d\lambda}.$$

The "pdf" of a first course is exactly this: a Radon–Nikodym derivative against length, existing precisely when $P \ll \lambda$. The yardstick is so universally taken for granted that no one notices the derivative is relative to anything — like asking "velocity with respect to what?" when everyone is standing on the same platform.

> **Insight** This is why the continuous KL-divergence could not be written as $\log(p/q)$: both point-masses are zero, so the ratio is $0/0$. Radon–Nikodym replaces mass-over-mass with *density-over-density against a common yardstick*,
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> $$\frac{p(x)}{q(x)} = \frac{dP/d\lambda}{dQ/d\lambda} = \frac{dP}{dQ},$$
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> and the arbitrary λ cancels top and bottom, leaving an intrinsic ratio of P to Q. The density formulation needs a yardstick; the ratio dP/dQ does not — which is the next regime's whole point.

Singular: continuous, yet no density exists

The split "discrete or continuous" is too coarse. There is a third kind: distributions with no atoms — every point still carries mass zero — that *also* possess no density. Put the uniform distribution on a circle in the plane $\mathbb{R}^2$. The circle carries all the probability, yet it has *area* zero, so $P \not\ll \lambda$ and no function f can satisfy $P(A) = \int_A f \, d\lambda$: the would-be density would have to be infinite along a one-dimensional curve and zero off it — a Dirac ridge, not a function. (On the line the classic specimen is the Cantor distribution: no atoms, no density.) The Lebesgue decomposition theorem — Radon–Nikodym's sibling — makes the taxonomy exhaustive: every

measure splits uniquely into an absolutely continuous part (it has a density), a singular part (it lives on a λ-null set, no density), and atoms.

Insight In the singular regime $dP/d\lambda$ and $dQ/d\lambda$ can *both* fail to exist, yet the direct ratio dP/dQ can still survive, provided $P \ll Q$ — the two measures living on the same thin set, compatibly. So the abstract form $D_{\mathrm{KL}}(P\|Q) = \int \log(dP/dQ)\, dP$ keeps working where the density form $\int p \log(p/q)\, d\lambda$ has nothing to stand on. That is not abstraction for its own sake; it is the only formulation that outlives the ambient density.

The hidden derivative in the language model

Now the connection the companion note left implicit.

The softmax is a Radon–Nikodym derivative — and it is what keeps the model safe

The vocabulary is a finite set of n tokens. Put the *counting measure* μ on it: every token has mass 1, so $\mu(A)$ is just the number of tokens in A. The model's output is a probability measure Q over tokens, produced by the *softmax* of the logits $z_w = \langle h, e_w \rangle$. That output is exactly the density of Q against counting measure:

$$\frac{dQ}{d\mu}(w) \;=\; q_w \;=\; \mathrm{softmax}_w\langle h, e_w\rangle \;=\; \frac{e^{z_w}}{\sum_k e^{z_k}}.$$

So the one-equation bridge of the companion geometry note *is* a Radon–Nikodym derivative: softmax is the discrete density, the dQ that every cross-entropy and divergence in these notes measures against. But two features of it are quietly doing the theorem's work for it.

It is strictly positive, so absolute continuity cannot fail. Because $e^z > 0$ for every finite logit, $q_w > 0$ for *every* token — Q has no null sets at all. That is not cosmetic: it forces $P_{\mathrm{data}} \ll Q$ *automatically*, so the one hypothesis that can ever fail — the absolute continuity that the Dirac mass and the circle were about — *cannot fail for a softmax model*. Equivalently, the loss $-\log q_y$ can never reach the infinity of an un-encodable symbol, because softmax never assigns a token probability exactly zero. The model is structurally incapable of declaring the observed token impossible.

It is the maximum-entropy form, so it is built from the entropy of the coding story. Over the simplex,

$$\mathrm{softmax}(z) \;=\; \arg\max_q \big[\langle q, z\rangle + H(q)\big], \qquad H(q) = -\sum_w q_w \log q_w,$$

the distribution that scores as high as possible on the logits while staying as high-entropy — as noncommittal — as it can, with the very entropy H of the coding story as the regulariser.

The exponential is forced here, not assumed. Maximising $\langle q, z\rangle + H(q)$ under the lone constraint $\sum_w q_w = 1$ has Lagrangian $\sum_w q_w z_w - \sum_w q_w \log q_w + \lambda\big(1 - \sum_w q_w\big)$; setting each $\partial/\partial q_w = z_w - \log q_w - 1 - \lambda = 0$ gives $\log q_w = z_w - 1 - \lambda$, so $q_w \propto e^{z_w}$, and normalising fixes the constant at $1/\sum_k e^{z_k}$. The only function whose logarithm is linear in the z_w is the exponential — that is where softmax's $\exp$ comes from. Positivity is automatic (an exponential is never zero), so the $q_w \geq 0$ constraints never bind, and the objective is concave on a convex set, so this stationary point is the unique global maximum. Phrased as a constrained problem instead — maximise $H(q)$ *alone*, subject to a fixed expected score $\langle q, z\rangle = c$ — the same computation returns $q_w \propto e^{\beta z_w}$, the Gibbs distribution, with the multiplier β (an inverse temperature) chosen to meet the constraint. Softmax is the flattest distribution consistent with a given average logit.

This is also why the loss is clean: $-\log q_y = \log \sum_k e^{z_k} - z_y$, and the log-partition $\log \sum_k e^{z_k}$ is the exponential-family normaliser whose gradient is the mean embedding $\bar{e}$ — the "spring law" of the geometry note.

Nobody needed to invoke Radon–Nikodym for any of this, because in the finite, strictly-positive case the theorem is empty: every positive function summing to one is a valid density, uniqueness is pointwise rather than almost-everywhere, and the hypothesis holds by construction. The theorem earns its keep only on continuous spaces; softmax is precisely what casts the discrete shadow.

The loss is an integral of the log-derivative

Here is the part that is not trivial. The note's per-example loss $L = -\log q_y$, summed over the corpus, is cross-entropy, and minimizing it is minimizing the Kullback–Leibler divergence between the empirical data distribution P_{data} and the model's distribution Q:

$$D_{\text{KL}}(P_{\text{data}} \,\|\, Q) \;=\; \int \log \frac{dP_{\text{data}}}{dQ}\; dP_{\text{data}}.$$

The integrand $\log \frac{dP_{\text{data}}}{dQ}$ is the logarithm of a Radon–Nikodym derivative. The divergence is only defined when $P_{\text{data}} \ll Q$ — the model must place positive probability on everything the data actually does. The note even writes a logarithm of such a ratio without naming it: its log-odds

$$z_i - z_j = \log \frac{q_i}{q_j}$$

are the logarithm of a density ratio between two tokens under the model's own distribution.

> The note says: "if the model has a low probability of predicting the correct next token then the loss is large." Measure theory says *why it must be*. If the model is not absolutely continuous over the data — if $q_y = 0$ for some token y that actually occurred — then $\frac{dP_{\text{data}}}{dQ}$ blows up and the KL divergence is infinite. Assigning zero probability to something that happened is the one unforgivable sin, and
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> $$L = -\log q_y \longrightarrow \infty \quad \text{as} \quad q_y \to 0$$
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> is that infinity showing through. The loss going to infinity and the derivative failing to exist are the same event in two languages.

> **Insight** So the geometry story and the measure story are one fact in two notations — exactly the move the companion note makes with "likely next word" versus "nearby direction." Here it is "cross-entropy loss" versus "integral of a log Radon–Nikodym derivative." Training a language model is performing a Monte Carlo estimate of $D_{\text{KL}}(P_{\text{data}} \| Q)$ and descending it, and absolute continuity is the hypothesis that makes the quantity it is descending finite in the first place.

The interesting direction: letting the vocabulary become continuous

The finite vocabulary keeps a language model in the first of the three worlds, where every token is a point with mass and the theorem sleeps. Replace the finite list with a continuum —

embeddings as a density on $\mathbb{R}^d$, or a diffusion model's score over a continuous state — and you fall straight into the third. The "meaning manifold" of the companion geometry note — a trained model concentrating probability on a lower-dimensional sub-manifold of the ambient space — is a *singular* measure: positive probability on a set of ambient measure zero, hence no density against Lebesgue. There is no $\mathrm{softmax}$-over-a-list to write: its denominator, a finite sum, becomes an integral $\int e^{z(x)}\,dx$ that need not even converge, and the resulting energy-based density need not exist against Lebesgue at all. Every guarantee softmax gave for free — a finite normaliser, strict positivity, automatic absolute continuity — becomes a genuine Radon–Nikodym question. The only thing that can still speak of one model's likelihood relative to another is the intrinsic ratio dP/dQ, which survives — by the previous section — exactly when the two models share the manifold. The "meaning manifold" and "the distribution Radon–Nikodym refuses a density" are the same phenomenon from two sides, and going continuous is what turns the silent foundation into the load-bearing one.